

SCOpt Using Higher-Order Unconstrained Binary Optimization (HUBO)

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1 Introduction

Abstract

This work presents a native Higher-Order Unconstrained Binary Optimization (HUBO) framework for global supply chain optimization using direct higher-order energy minimization without quadratization. Unlike traditional QUBO formulations that reduce higher-order interactions into pairwise couplings through ancilla-variable expansion, the proposed framework preserves native cubic logistics interactions and optimizes them directly using classical simulated annealing. The system models supplier-demand routing, transportation costs, activation dynamics, and multi-way logistical synergies as sparse higher-order tensor interactions.

The proposed formulation avoids topology distortion, ancilla proliferation, and coefficient inflation typically introduced during quadratization. Instead, the optimization problem is represented directly as a sparse higher-order energy landscape over binary shipment variables.

2 Problem Formulation

Consider a logistics network consisting of:

- N_s supply sites
- N_d demand sites
- A single part type
- Sparse transportation connectivity

Each binary decision variable is defined as:

$$X_{ij} \in \{0, 1\} \tag{1}$$

where $X_{ij} = 1$ indicates that supplier site i ships inventory to demand site j .

3 Objective Function

The HUBO energy function is defined as:

$$H = H_{\text{transport}} + H_{\text{inventory}} + H_{\text{demand}} + H_{\text{higher-order}} \quad (2)$$

3.1 Transportation Cost Term

$$H_{\text{transport}} = \sum_{i,j} C_{ij} X_{ij} \quad (3)$$

where C_{ij} denotes the transportation cost from supplier i to demand destination j .

3.2 Inventory Constraint Penalty

$$H_{\text{inventory}} = P_1 \sum_i \left(\sum_j X_{ij} - I_i \right)^2 \quad (4)$$

where:

- I_i is the available inventory at supplier i
- P_1 is a soft penalty coefficient

3.3 Demand Satisfaction Penalty

$$H_{\text{demand}} = P_2 \sum_j \left(\sum_i X_{ij} - D_j \right)^2 \quad (5)$$

where:

- D_j is the demand at destination j
- P_2 is a demand penalty coefficient

3.4 Higher-Order HUBO Interaction

A native cubic HUBO interaction is introduced to model multi-way logistical synergy:

$$H_{\text{higher-order}} = \lambda \sum_{a,b,c} Y_a H_b F_c \quad (6)$$

where:

- Y_a represents supplier activation
- H_b represents logistics hub activation
- F_c represents freight carrier utilization

This cubic interaction directly captures higher-order logistical dependencies without quadratization.

4 Sparse Variable Construction

To reduce combinatorial explosion, only physically feasible shipment variables are constructed. Variables are generated only if:

$$I_i > 0 \quad \text{and} \quad D_j > 0 \quad (7)$$

This eliminates impossible logistics states from the search space prior to optimization.

5 Native HUBO Optimization

Unlike QUBO solvers restricted to pairwise couplings, the proposed framework directly evaluates arbitrary-order tensor interactions during annealing. The total energy is computed as:

$$H = \sum_t w_t \prod_{v \in t} x_v \quad (8)$$

where:

- w_t is the coefficient of HUBO term t
- x_v is the binary state of variable v

The optimizer performs direct higher-order energy minimization using classical simulated annealing.

6 Advantages of Native HUBO Optimization

The proposed approach offers several advantages over quadratized QUBO systems:

- **No ancilla-variable expansion:** Avoids exponential expansion of problem dimensions.
- **Preservation of higher-order interaction topology:** Maintains natural structural constraints.
- **Reduced variable count:** Lower memory footprint and smaller search spaces.

- **Improved sparsity:** Maximizes computational acceleration paths.
- **Elimination of quadratization penalties:** Prevents coefficient range inflation and numerical instability.
- **More physically meaningful energy landscapes:** Enhances structural path tracing during optimization.

This framework is particularly suitable for large-scale enterprise logistics systems containing naturally occurring multi-way interactions.

Objective Function HUBO

Decision Variables

$$X_{ijk} \in \mathbb{Z}_{\geq 0}$$

Units of part k transported from source site i to destination site j .

$$T_{mjk} \in \mathbb{Z}_{\geq 0}$$

Units of part k newly procured from supplier m for destination site j .

$$U_{jk} \in \mathbb{Z}_{\geq 0}$$

Unmet demand of part k at destination site j .

$$Y_i \in \{0, 1\}$$

Binary activation variable indicating whether source site i ships inventory.

$$Z_j \in \{0, 1\}$$

Binary activation variable indicating whether destination site j receives inventory.

$$F_c \in \{0, 1\}$$

Binary variable indicating whether freight carrier c is utilized.

$$H_h \in \{0, 1\}$$

Binary variable indicating whether logistics hub h is activated.

$$G_r \in \{0, 1\}$$

Binary variable indicating whether global/tariff route r is activated.

Structural Feasibility Constraint

$$X_{ijk} = 0 \quad \forall (i, k) \text{ such that } E_{ik} = 0$$

A site with no excess inventory for part k cannot act as a source node for that part.

Objective Function

The HUBO objective function is defined as:

$$\begin{aligned}
H = & \sum_{i,j,k} C_{ijk} X_{ijk} + \sum_{m,j,k} B_{mjk} T_{mjk} + UD \sum_{j,k} U_{jk} \\
& + FC_1 \sum_i Y_i + FC_2 \sum_j Z_j \\
& + P_1 \sum_{i,k} \left(E_{ik} - \sum_j X_{ijk} - S_{ik} \right)^2 \\
& + P_2 \sum_{j,k} \left(N_{jk} - \sum_i X_{ijk} - \sum_m T_{mjk} - U_{jk} \right)^2 \\
& + P_3 \sum_i \left(\sum_{j,k} X_{ijk} - MY_i \right)^2 \\
& + P_4 \sum_j \left(\sum_{i,k} X_{ijk} + \sum_{m,k} T_{mjk} - MZ_j \right)^2 \\
& - \lambda_1 \sum_{i,h,c} Y_i H_h F_c \\
& + \lambda_2 \sum_{r,c,j} G_r F_c Z_j \\
& - \lambda_3 \sum_{i,j,k_1,k_2,k_3} X_{ijk_1} X_{ijk_2} X_{ijk_3} \\
& + \lambda_4 \sum_{a,b,c} Y_a Y_b Y_c
\end{aligned}$$

Objective Function Components

- C_{ijk} : transportation cost for shipping part k from site i to site j
- B_{mjk} : procurement cost for purchasing part k from supplier m for destination j
- UD : unmet demand penalty coefficient
- FC_1 : fixed source-site activation cost
- FC_2 : fixed destination-site activation cost
- P_1 : inventory conservation penalty coefficient
- P_2 : demand satisfaction penalty coefficient

- P_3 : source activation consistency penalty
- P_4 : destination activation consistency penalty
- λ_1 : central hub and freight synergy reward
- λ_2 : tariff/global routing interaction penalty
- λ_3 : shipment consolidation reward coefficient
- λ_4 : supply-chain resilience/risk penalty coefficient